Eliminate Every Obstacle Between Your Data and Your GPUs

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Agents demand a new paradigm for data retrieval

TRADITIONAL APPLICATIONS

Stateless requests

Per-request transaction, short-lived

Structured data

SQL tables, rows, well-defined schema

One workflow, one user

Request → response

Data at rest, data in motion

Well-defined workflows

AGENTIC AI SYSTEMS



Long-running sessions

Hours to days, sub-agents, tool chains



Unstructured + structured

Docs, PDFs, video, audio, tables, streams



Hundreds to thousands of agents

Per tenant, per workflow, concurrent



Context is new category of primary data

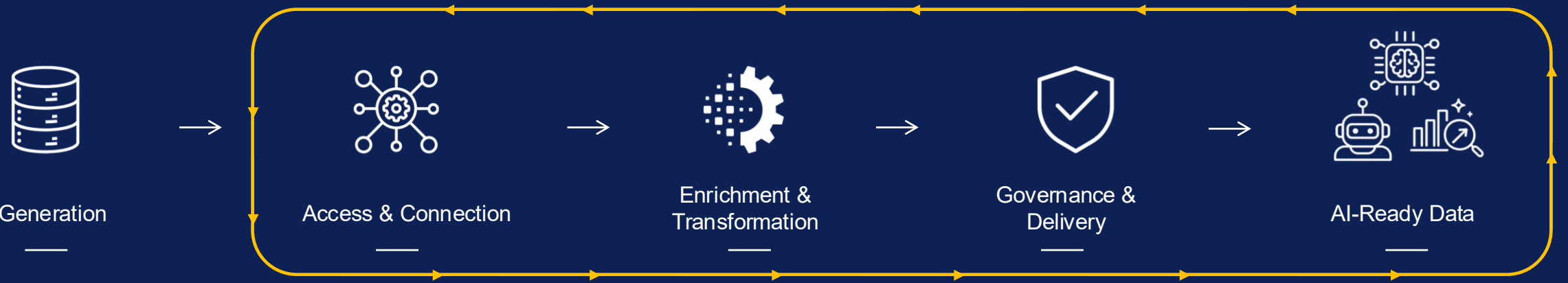
KV cache, embeddings, memory, scratch

How raw data becomes AI-ready

Physical World | Crude Oil Refining



Digital World | AI Data Refining



AI DATA PLATFORM

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Dell AI Data Platform

Turn siloed data into one enterprise-wide context layer

Ingest | Prepare | Enrich | Deploy

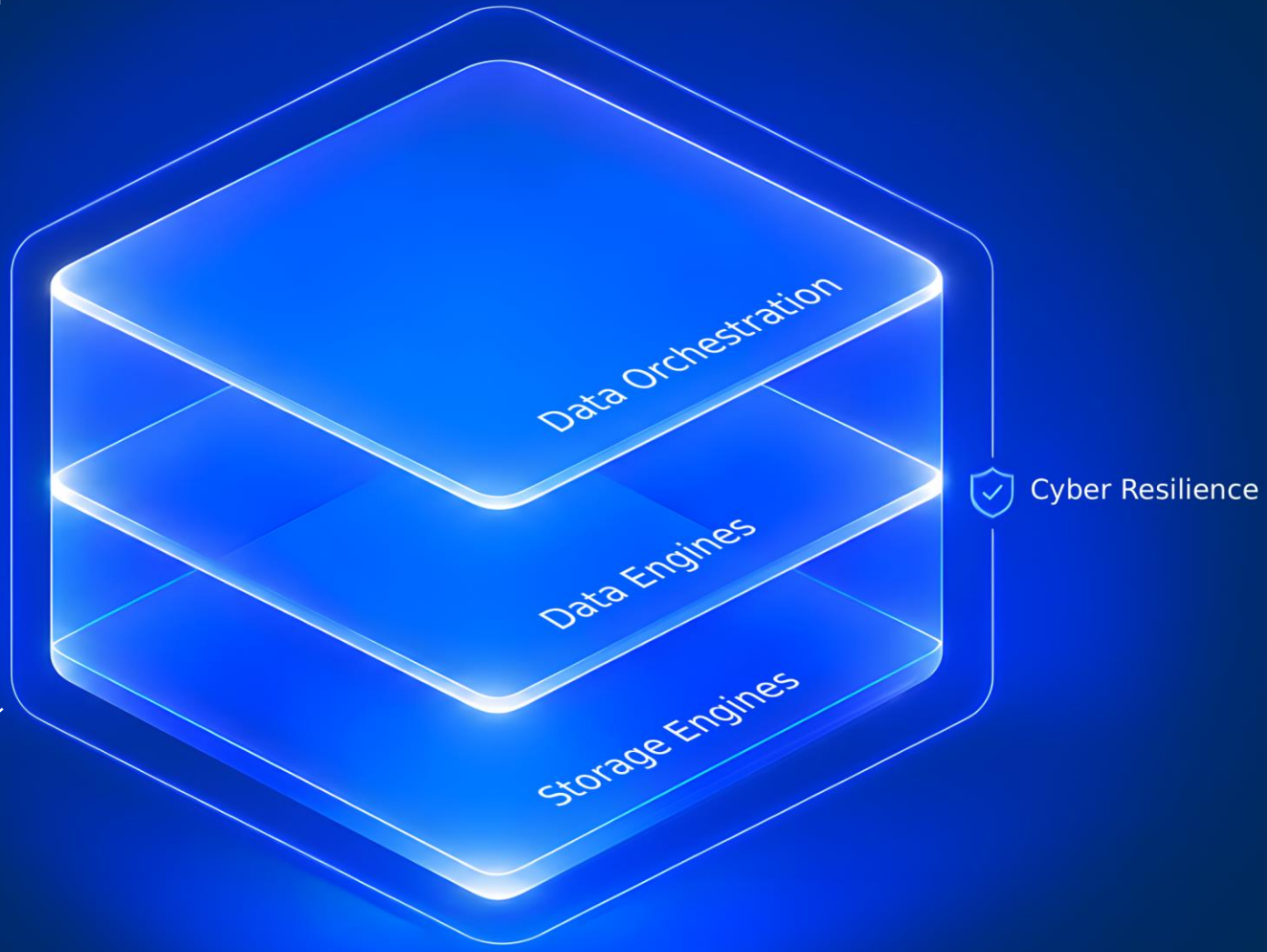
Deploy agents and models at scale - with guardrails

Processing | Analytics | Search

Reduce data discovery from weeks to hours

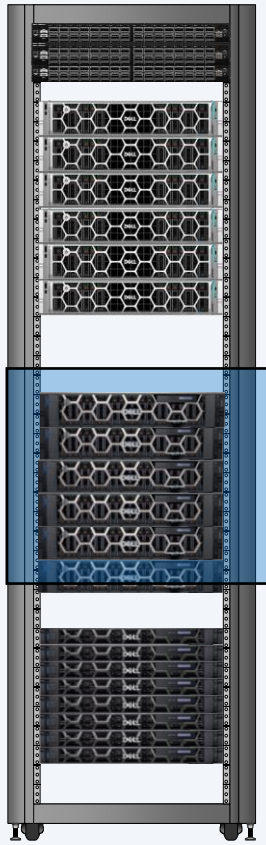
Scalable | Secure | Performant

Never starve another GPU again with best-of-breed AI storage



Dell AI Factory: Where Compute, Storage, and Data Converge

Dell AI Factory



Networking

AI Compute (PowerEdge R7725 | XE7745)

Storage Engines (PowerScale, ObjectScale, PowerFlex, Lightning FS)

Data Engines +Data Orchestration (including Search, Analytics, Processing)

Agentic AI

Runs orchestration models and multi-agent inference at scale

Stores raw data, agent state, logs, KV cache of LLM tokens, and tool outputs

Orchestrates ETL, embedding generation, and vector indexing across structured and unstructured data

Dell Professional Services

Strategize • Implement • Adopt • Scale

Three obstacles between data and GPUs

Too many AI deployments stall before they ever start.

GPU utilization

File, object, block and parallel workloads each require different storage outcomes — and agentic inference has outgrown GPU memory. Get any of it wrong and your GPUs are not utilized to their potential

Storage choices lock you in

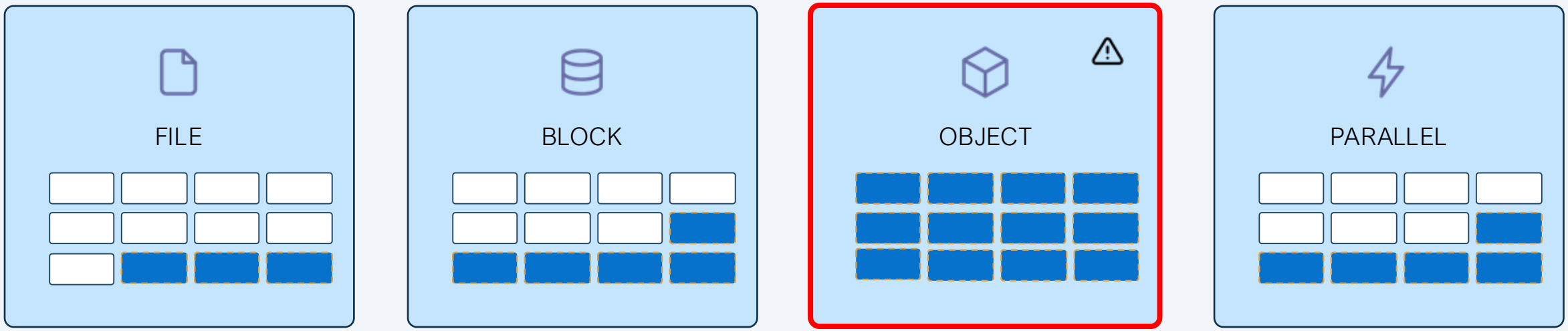
Buy a silo per workload and capacity strands where it sits. Buy one “unified” system and every workload competes for resources and performance, often through additional abstraction layers.

Integration burns months

Compute, networking and storage arrive from separate vendors, on separate contracts and timelines — all before a single model is deployed.

How To Efficiently Run File, Block, Object, Parallel File Simultaneously

If storage is strained, GPUs sit idle



The Problem: Overprovisioned - Idle Capex - Siloed

At scale, ‘unified’ means compromise

A single code path must pick a winner on every axis — these are opposing designs, not tuning knobs

	BLOCK	FILE	OBJECT
✕ SEMANTICS	Ordered sync writes	POSIX in-place edits	Immutable PUT
✕ ALLOCATION	4–64 KB aligned	Variable extents	MB-scale stripes
✕ METADATA	Tiny map, sub-ms	Inodes · ACLs · trees	Flat index, billions
✕ PROTECTION	Mirror / narrow RAID	Mixed profiles	Wide erasure coding
✕ LOCKING	Host-side writer	Byte-range locks	None — last wins

Fundamental breaks

- ✕ One allocator, three optimizations
- ✕ One index, opposing scales
- ✕ One layout, one preferred protocol

THE EMULATION TAX | one native protocol plus gateways — the rest inherit latency or lose semantics

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Dell's Best-of-breed Storage Engines

PowerScale

Unified file and object storage from AI to archive



- Multiprotocol namespace
- RDMA access
- pNFS, Fully scalable data and metadata service
- Zero-Trust + cyber vault

ObjectScale

Object storage at enterprise scale, from AI to archive



- Millions of IOPS, sub-ms
- Linear scale 1,000+ nodes
- Multi-tenancy & QoS
- Erasure Coding

Lightning FS

Extreme AI training and inferencing



- Exabyte-scale S3
- Global namespace
- Native S3 Tables and Vectors
- Object Lock compliance

PowerFlex

Enterprise scale-out block for databases



- 150 GB/s per node
- Cache-free, zero tuning
- Tier 0 for extreme inference and training performance
- Direct Client Data Access via NVME/RDMA
- Scalable MetaData Servers

Available now as a single **Dell Exascale Storage** offer

Coming soon

Exascale: One Storage Platform - All your Data.

Four stacks each bring their own strengths. Exascale is all of them at once

Best-of-breed software

- Parallel File System
- Block — COMING SOON
- Object
- File

The stacks stay purpose-built. The fleet beneath them converges

PowerEdge server foundation engineered for storage software

Compounding efficiency across purchasing, sparing, operation, & re-use



Software-defined by design

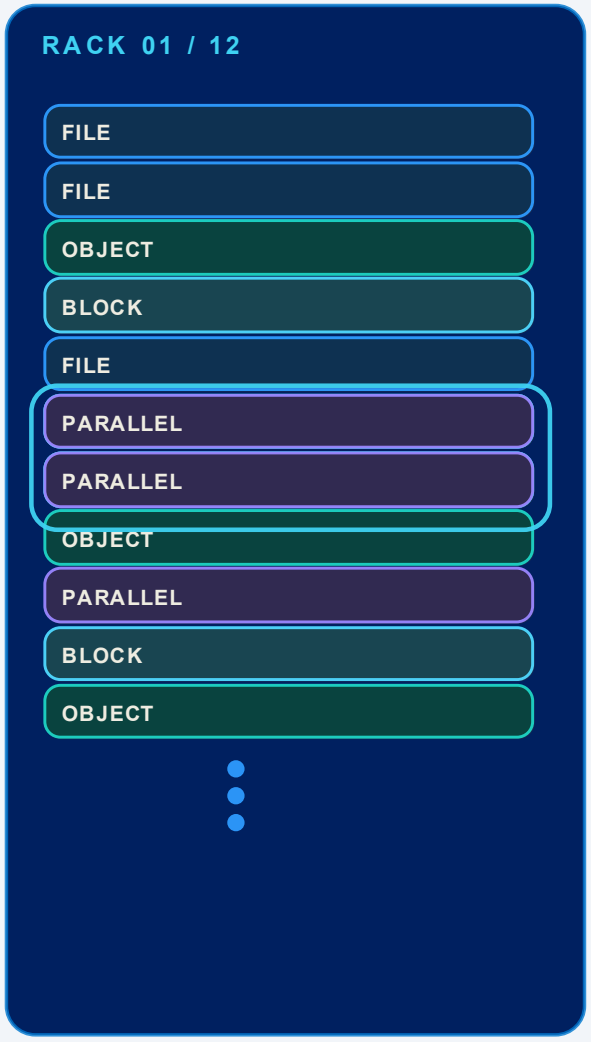
One license turns common PowerEdge hardware into file, object, parallel file, or block.

Single License Agreement

One predictable per-terabyte agreement that adapts as your software mix changes.

Compose by the node

One personality per server — provision, then re-provision in place.



PERSONALITIES

- PowerScale · File
- ObjectScale · Object
- Lightning Filesystem · Parallel File
- PowerFlex · Block

SEQUENCE

1 · Provision — assign one personality to every node

2 · Re-provision in place — PowerScale ▸ ObjectScale

3 · Re-provision again — ObjectScale ▸ Lightning Filesystem

AVAILABLE TODAY

R7725xd

2U · dual socket



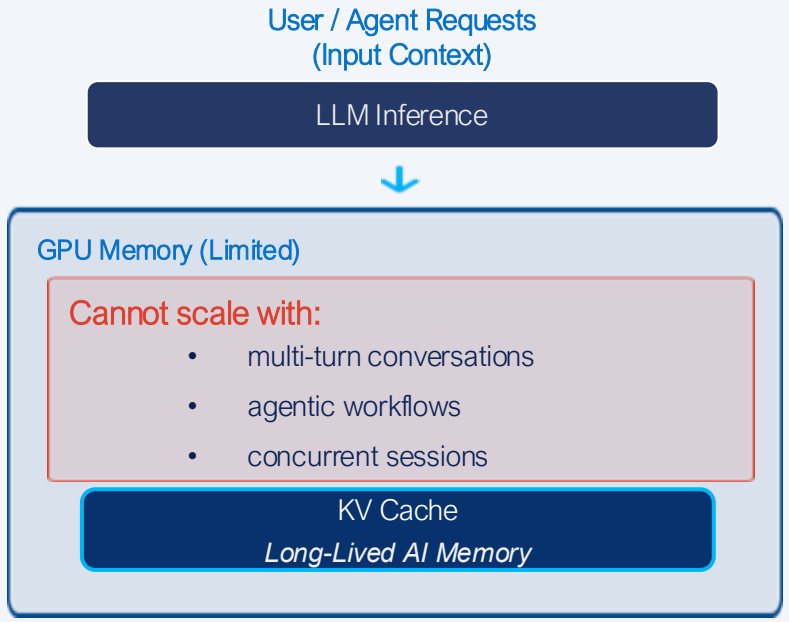
24 × NVMe

CPU	MEMORY	PER DRIVE	NETWORK
Dual AMD EPYC	768 GB	7–245 TB	4 × 400GbE

Compose at rack scale: mix node personalities, then re-provision any node on demand — same hardware, no forklift.

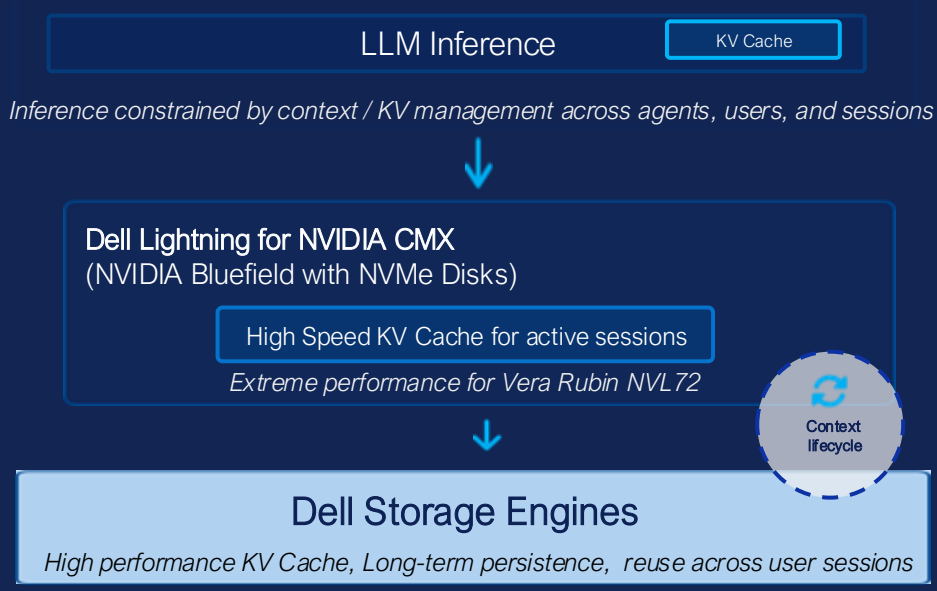
Inference at Scale: AI Has Outgrown GPU Memory

Delivering efficient inferencing in the agentic era



- Every Prompt generates KV tensors that grow in length with each token.
- Transforming input context is GPU compute intensive
- Agentic and multi-step workflows require persistent KV state.
- GPU and CPU memory alone can not scale to meet this demand.

Inference is **constrained by context / KV** management across agents, users, and sessions – not compute



Offloading KV Cache to Dell Storage

- **Burns fewer GPU cycles** — context computed once, not recomputed every turn
- **Scale to longer contexts and more concurrent users** — without adding GPUs
- **GPUs freed for higher-value work** — cycles not wasted on recomputation

Read More: Dell Storage Engines: Accelerating AI inferencing
Faster GPUs and Smaller Caches Raise the Bar for Storage

Racked, Stacked, and Ready to Run

#1 rack-scale infrastructure provider in CY2025[1]

Dell PowerRack delivers compute, networking and storage as one pre-validated, factory-integrated blueprint

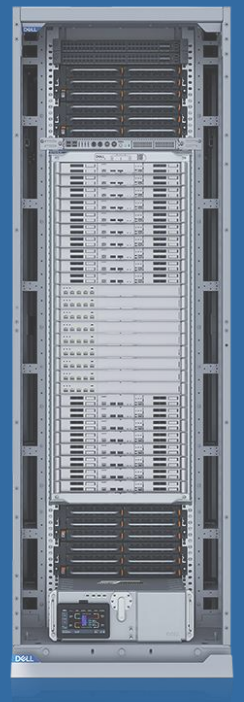
Competitors

- Compute
- Network
- Software
- Support
- Storage

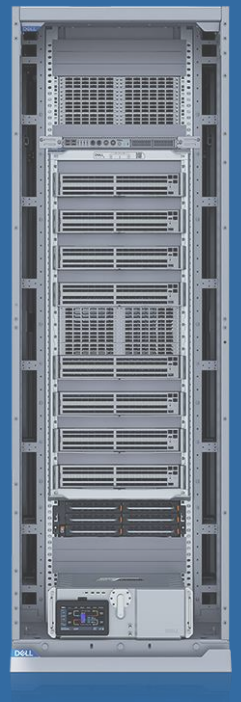
Competitors require multiple vendors, multiple contracts, multiple timelines — all before you earn a dollar.

Dell PowerRack for compute, networking and storage

PowerRack for compute



PowerRack for networking



PowerRack for storage



Assess & size

Right-sized from day one

Design & validate

Pre-validated for GPU density and throughput

Build & integrate

Factory-assembled before it ships

Deliver & deploy

6 hours from delivery to production[2]

1. IDC Quarterly Server Tracker Q4 2025 - CLM-016432 - Dell is the #1 rack scale infrastructure provider in CY2025, leading the industry in both units shipped and revenue share.
2. Based on a Principled Technologies report commissioned by Dell, Accelerate AI time to value with Dell Services, May 2026* Testing reflected an air-cooled single rack 16-GPU AI Factory solution. Actual results may vary

Leverage Dell's Global Supply Chain to Deliver

- 5,000+ experts focused solely on global supply-chain optimization.
- Among the largest buyers of GPUs, CPUs, memory, and flash in the world.
- #1 in client, server, and storage — giving Dell unmatched leverage with suppliers.
- Proven global powerhouse for reliability, resilience, and execution.

One of the
Largest
GPU purchasers in the world

#1
in client, server and storage

One of the
Largest
purchasers of CPU, memory and flash in the world

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So why choose the Dell AI Data Platform?

Delivering AI Storage that eliminates obstacles

Storage needs to do more than just retain data: it needs to offer a reliable layer to connect context to action.

Maximizing GPU Utilization

Storage Engines match the data path to the workload, meaning GPUs are never waiting for data. And, KV Cache offloading means GPUs are highly utilized doing net new work

No lock-in

Exascale runs every storage engine on common storage-optimized PowerEdge hardware. Re-provision nodes in place as workloads shift — no forklift, no new silo, no stranded capacity

Ready to run

PowerRack arrives factory-integrated and pre-validated — compute, networking and storage, racked and stacked — from the #1 rack-scale infrastructure provider.[1]

[2] IDC Quarterly Server Tracker, Q4 2025: Dell is the #1 rack-scale infrastructure provider in CY2025, in units shipped and revenue share.